

How a thin market works, and how to sell \$1,000,000 equivalent into it

This study examines the market microstructure and market-maker behaviour of a thinly traded token, iExec (RLC), and draws out what they imply for liquidating a large position.

8 June 2026

THE PROBLEM

The task is to sell \$1,000,000 of RLC, a thinly traded token in a sustained decline



The task

The objective is to liquidate a large position for net proceeds of **\$1,000,000 USDC**. The sale must be executed in the spot market, entirely on public exchanges, with no recourse to over-the-counter (OTC) block trades, and at the lowest achievable cost.

Two difficulties compound. The market is thin, and the asset is falling. The analysis therefore proceeds in two stages: it first establishes how the market works, and then derives what that structure implies for the sale.

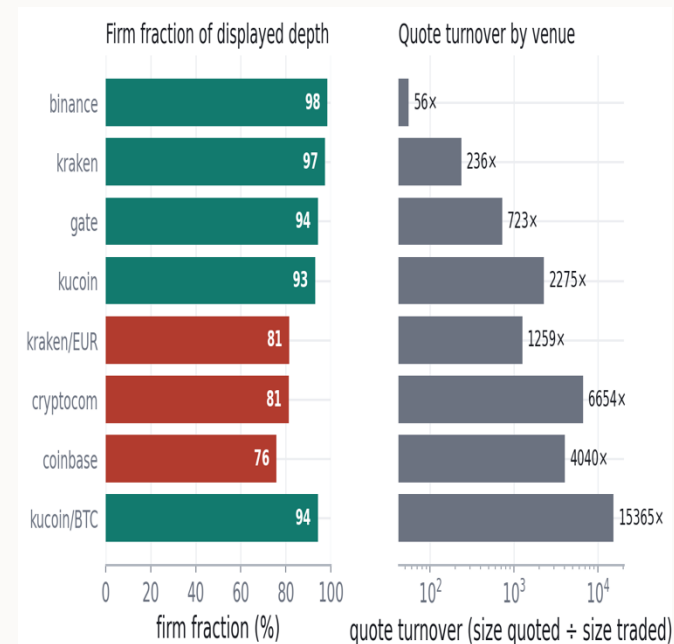
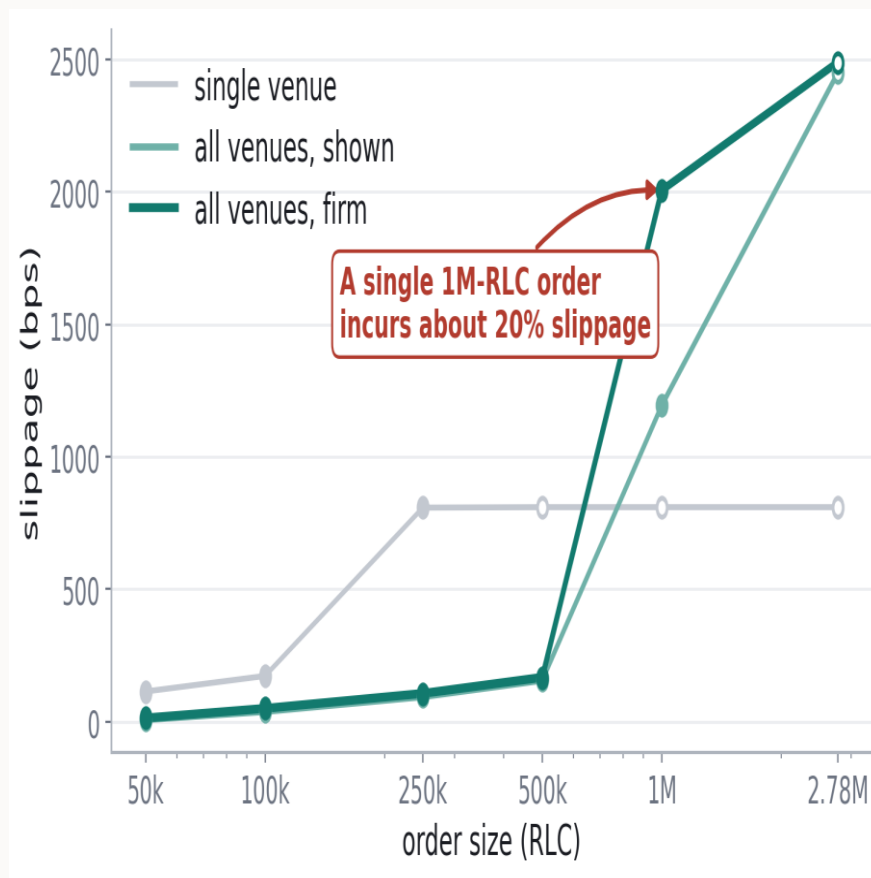
PART ONE

How the market works

This part describes the market's structure: how thin it is, who provides liquidity on the other side of the trade, and how the venues move in relation to one another.

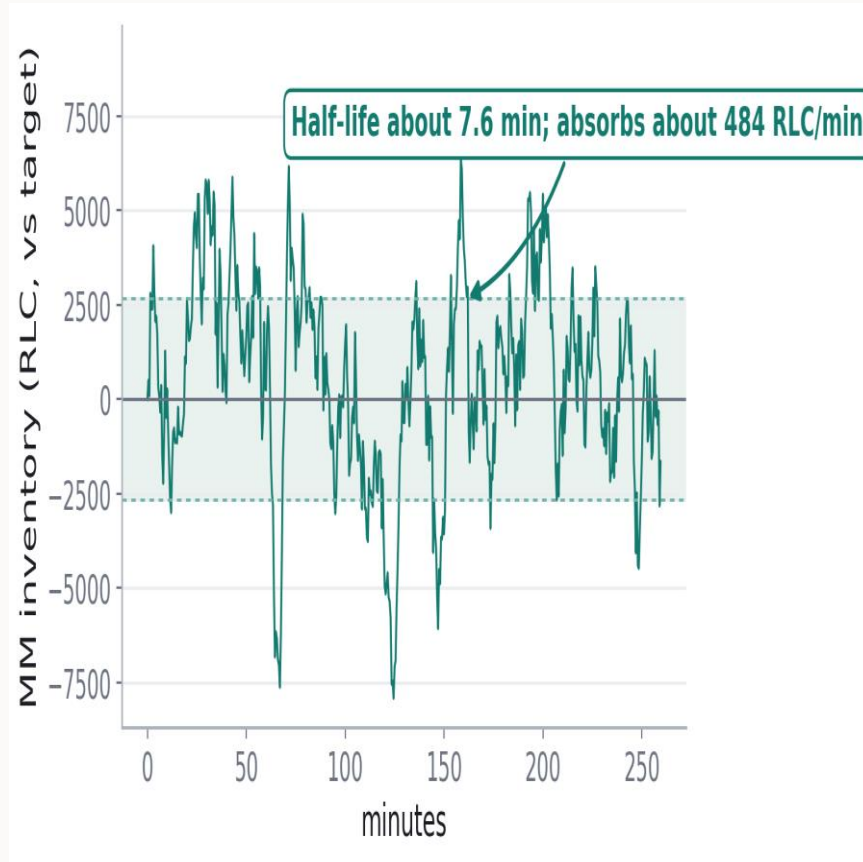
LIQUIDITY

The order book is thin, expensive to trade in large size, and much of its displayed depth is not firm



In summary. The cost of trading stays modest up to a few hundred thousand RLC and then rises steeply. The venue that displays the most depth, Coinbase, is only 76% firm: for every unit actually traded at the best quoted prices, roughly four thousand units are posted and then cancelled. Dependable depth is therefore small. Firm depth is the size that rests long enough to be executed against; the rest disappears before an order can reach it.

The liquidity providers on the other side absorb only about 484 RLC per minute before they withdraw their bids

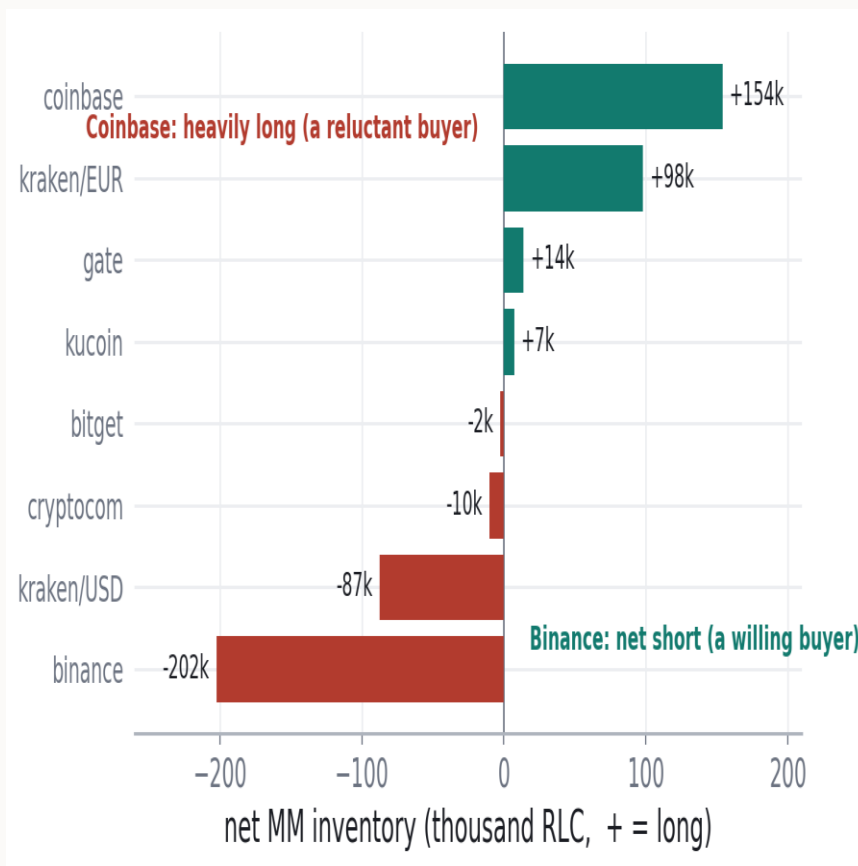


The mechanism

Market makers do not hold the tokens they buy. They carry inventory only briefly and steadily return it toward a target level. Modelling this as a mean-reverting process gives a rebalancing half-life of about 7.6 minutes and a **sustainable absorption rate of roughly 484 RLC per minute**. That rate is the effective limit on how quickly the position can be sold.

A second constraint. Market makers widen their quoted spreads when volatility rises, a defence against trading with better-informed counterparties. Selling aggressively in volatile periods therefore costs more.

Dealer positioning is uneven across venues: Binance market makers are short, while Coinbase market makers are heavily long

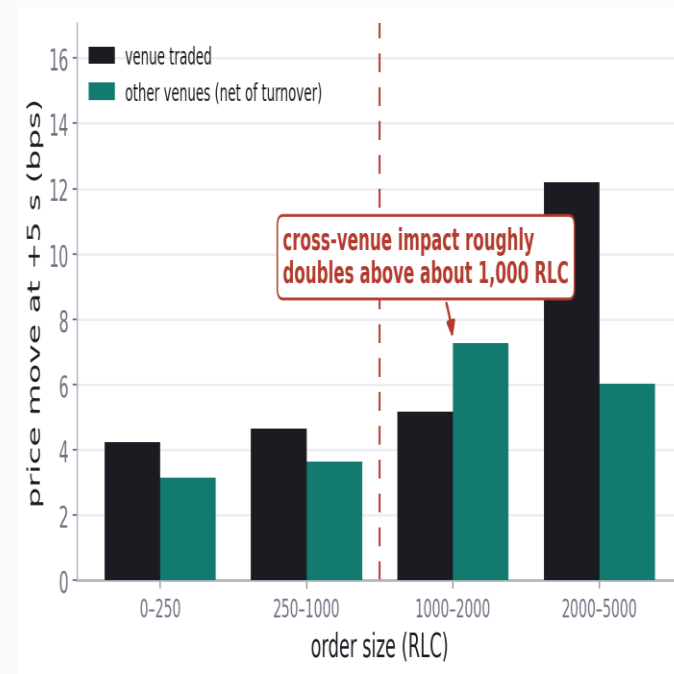
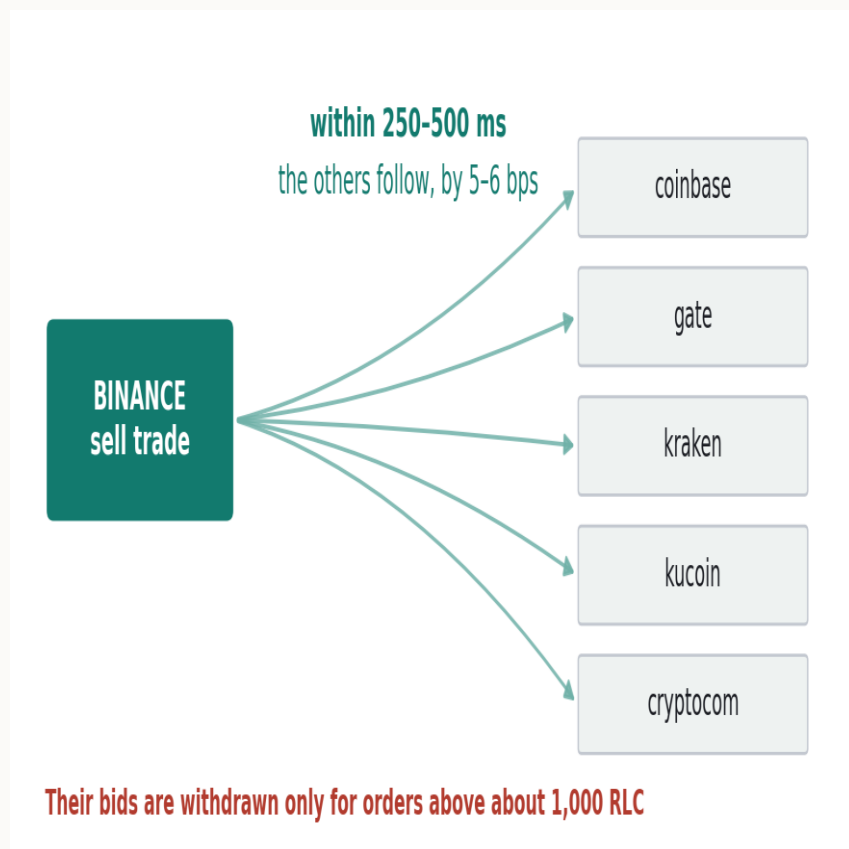


Why the choice of venue matters

A market maker who is short wishes to rebuild inventory and is therefore a willing buyer of the tokens being sold. One who is already heavily long is reluctant to take on more. The choice of venue, and of counterparty, is consequently not a matter of indifference.

An important caveat. The absolute level of inventory cannot be identified from the data and shifts with the estimation method. Only the differences in positioning across venues and the absorption rate are robust.

The six venues move together: a trade on Binance is reflected on the others within roughly half a second



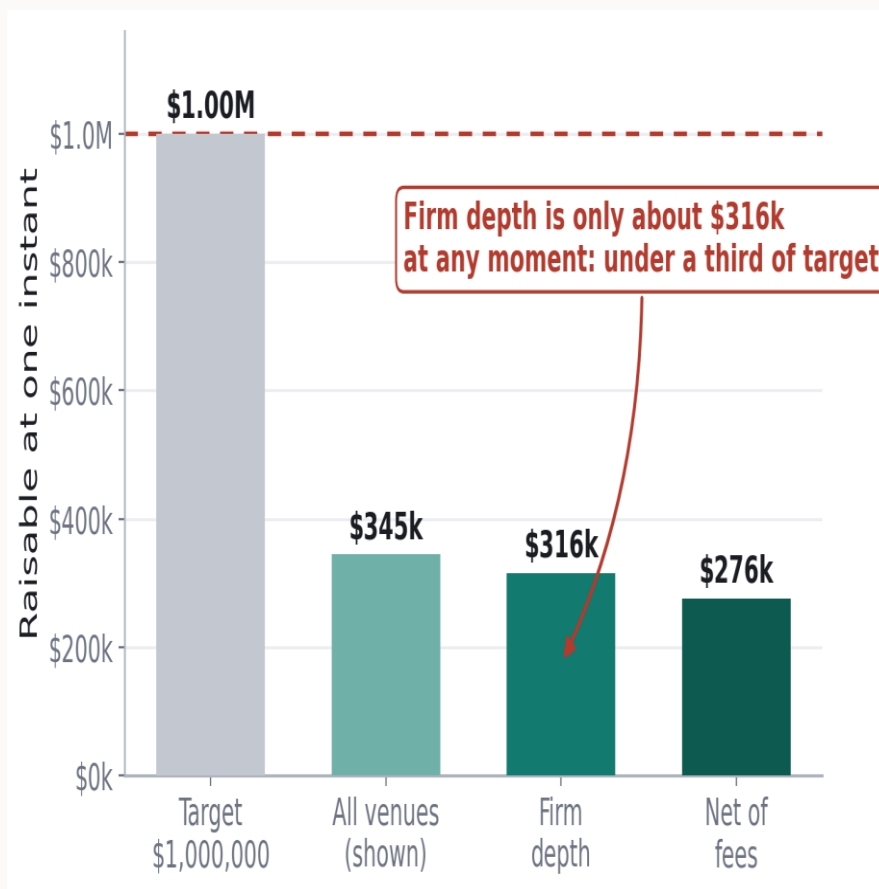
The implication for order size. An order below about 1,000 RLC is absorbed within the normal turnover of quotes and leaves little trace. Above that size, the impact that propagates to the other venues roughly doubles. Individual orders should therefore be kept small.

PART TWO

What this implies for the sale

The structure of the market determines the liquidation directly: the limit on how fast it can proceed, the trade-off between speed and cost, and the resulting execution program.

At any moment the market holds only about \$316,000 of firm depth, roughly one third of the target

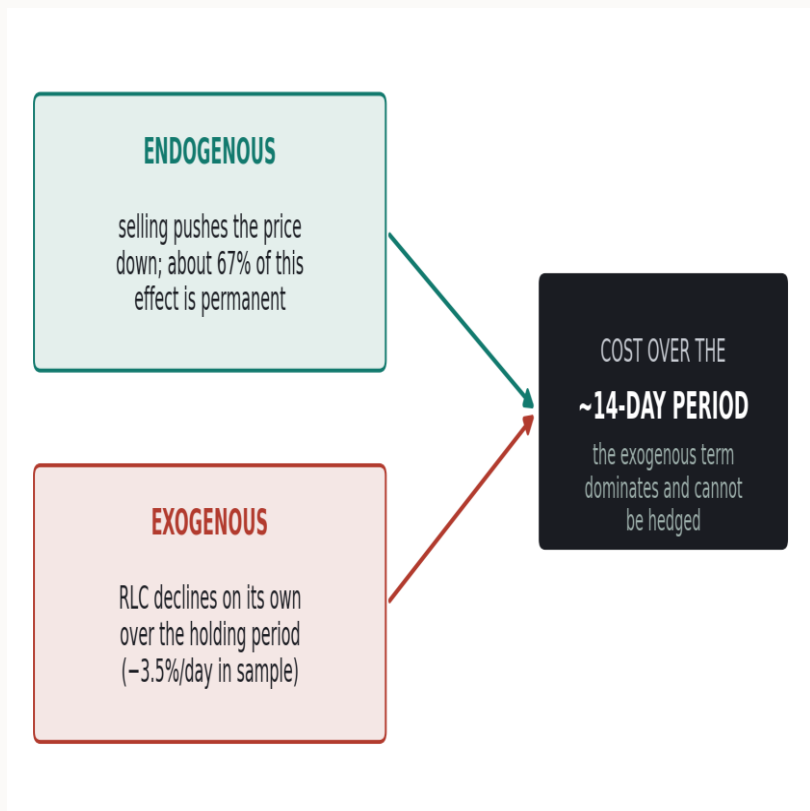


Why this is a question of time

The target of \$1,000,000 corresponds to about **2.9 million RLC**. That is roughly three and a half times a full day of genuine trading volume, and about three times the entire firm depth available across all venues. Neither a single moment nor a single day can absorb it.

It follows that the sale must be spread over time, which in turn exposes it to movements in RLC's price. That trade-off is the subject of the next section.

The cost separates into two components, and the one that cannot be controlled is the larger

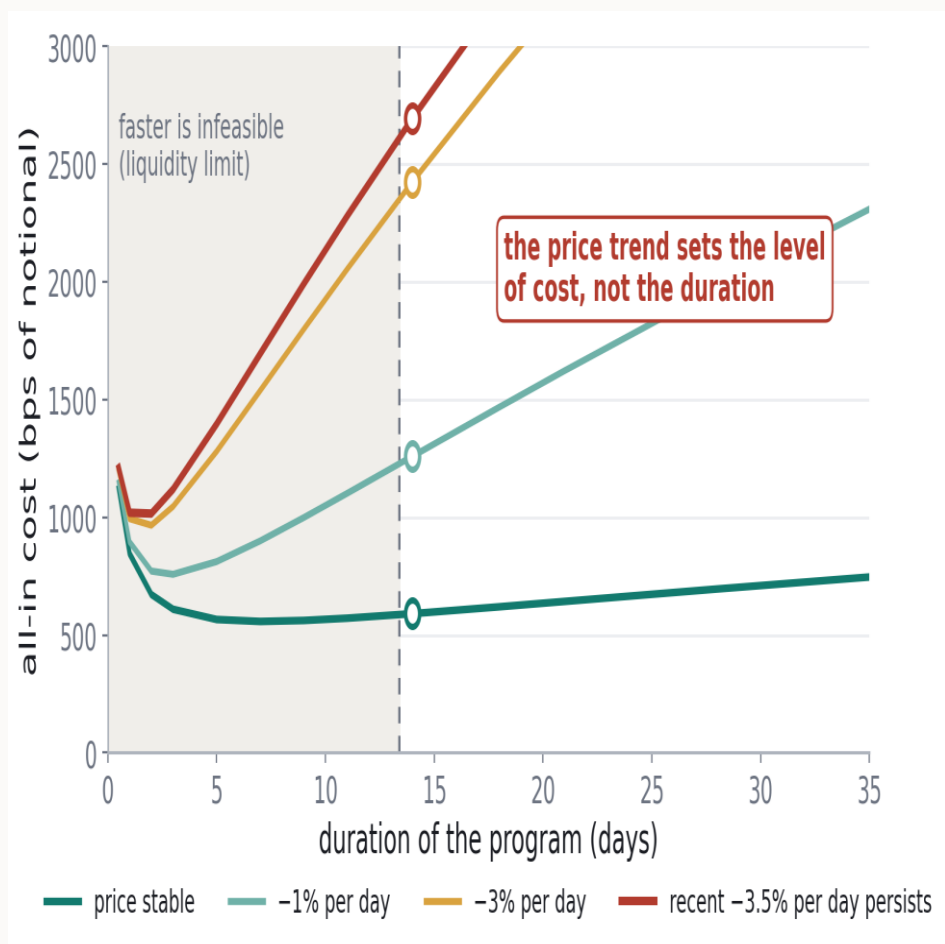


The endogenous component is the downward pressure created by the act of selling. Part of this effect is temporary and part is permanent.

The exogenous component is RLC's own decline over the period during which the position must be held.

The central point. Execution can be optimised, but the underlying price trend cannot. The trend accounts for most of the cost.

Selling more slowly raises exposure to the decline, and selling faster is infeasible: the floor is about thirteen days



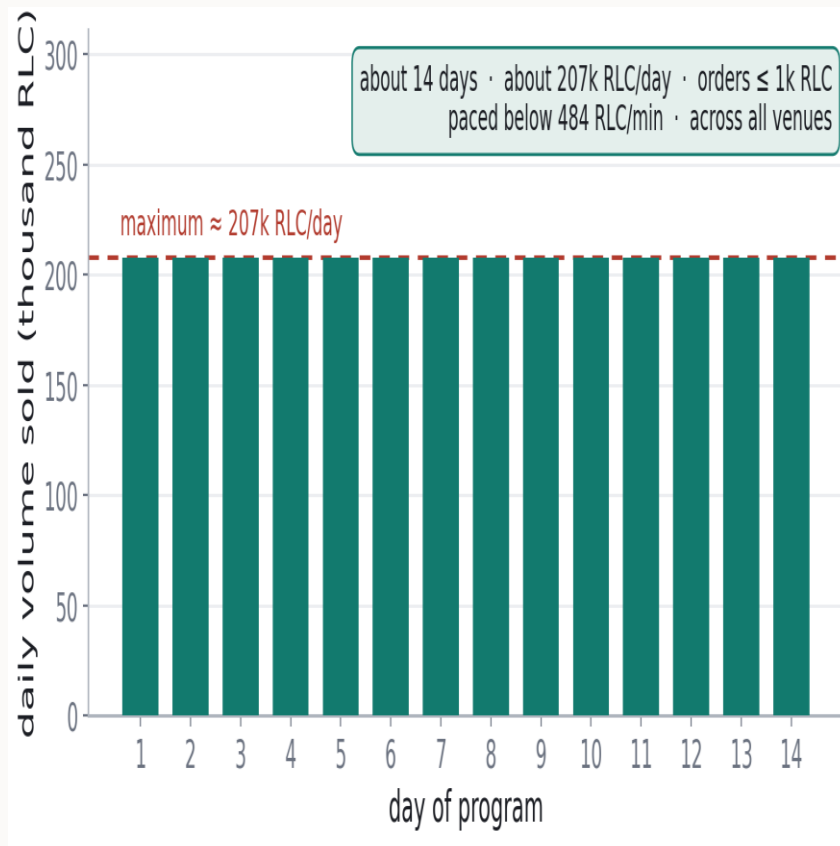
How to read the frontier

The shaded region marks durations that liquidity rules out, since nothing faster than about thirteen days can be achieved. Beyond that limit, a longer schedule only adds exposure to RLC's decline. The four curves correspond to different price scenarios and run nearly parallel, which shows that the trend sets the level of cost rather than the best duration. **The implied choice is to trade at the feasible floor of about thirteen to fourteen days.**

An independent check. A square-root impact law that uses no order-book data places the impact term at about 152 basis points, consistent with the frontier.

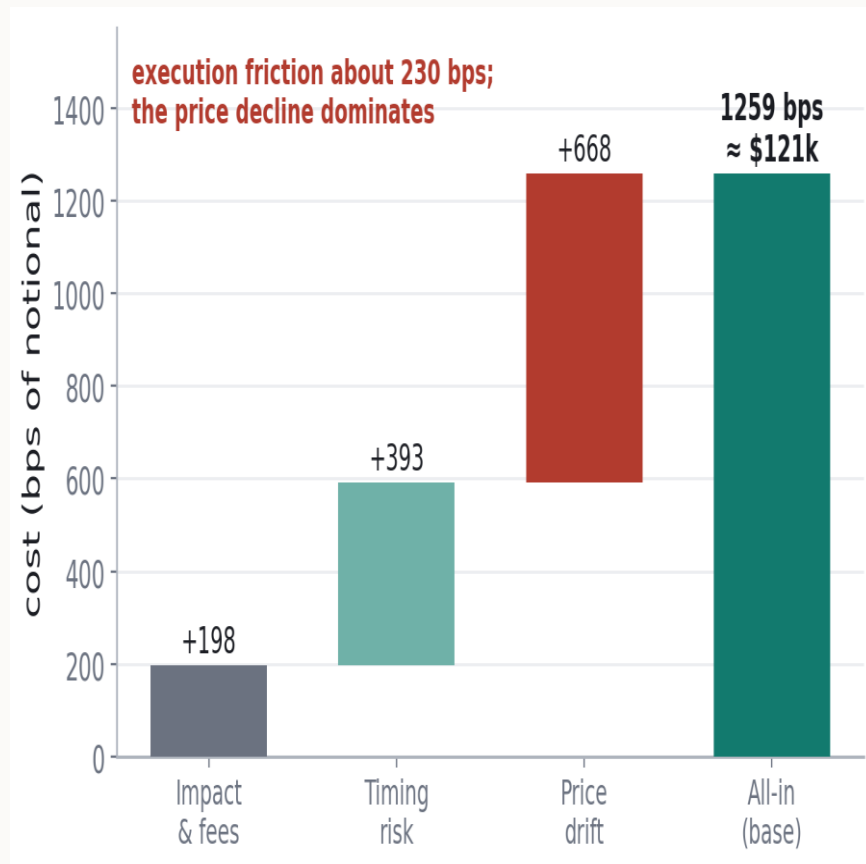
THE PROGRAM

Execute the full position on public venues, at about 207,000 RLC per day for roughly fourteen days, in small orders across all venues



- 1 Trade at the maximum sustainable rate.** About 207,000 RLC per day; because this rate is the binding constraint throughout, the program should run at it from the first day rather than building up to it.
- 2 Use small orders at a measured cadence.** Each order should be no more than about 1,000 RLC, paced below 484 RLC per minute and spaced to let the order book replenish.
- 3 Distribute across the venues.** Choose venues according to available depth, fees, and dealer positioning, and avoid revealing the full intention on any single exchange.
- 4 Concentrate in the deepest hours.** Weight execution toward the most liquid windows, around 07:00 and from midday to evening UTC.

Execution friction accounts for only about 230 basis points; the larger cost is the price decline borne while holding



Base case: a decline of 1% per day

Impact, fees, and cross-venue coupling together come to **about 230 basis points**.

The timing risk over the two-week holding period is **large, near $\pm 13\%$ at one standard deviation**.

The exogenous price decline is **the dominant term, and it depends on the scenario**.

The decisive lever. The aim is not to reduce execution slippage by a few basis points, but to minimise the number of days the position is held. The market itself sets the floor on that number.

What the analysis cannot establish

Drift is an estimate, not a forecast.

The decline of about 3.5% per day drives the cost band, but five days of data cannot predict the next two weeks. The result is reported as a band rather than a single figure.

Large-order impact is bounded, not measured.

The trades observed in the sample are small. The scaling of impact at execution sizes is reported as a range tested across plausible values and cross-checked against the square-root law.

Dealer inventory is inferred.

The data contain no order identifiers. The level of inventory and any switching point cannot be identified; only the absorption rate and the dispersion across venues are robust.

The sample is short and covers a single regime.

The window spans about 4.7 days, declining and then flattening. It covers spot markets only, and hidden depth and position in the order queue are not observable.

What to do, and what to monitor

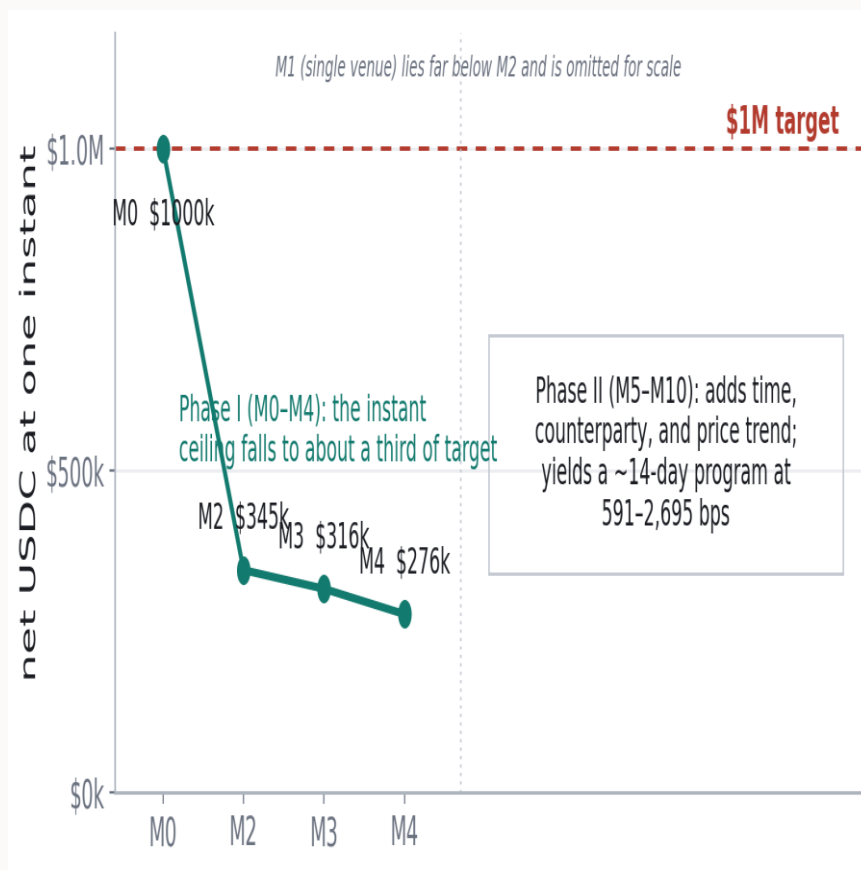
- 1 This is a program, not a single trade.** \$1,000,000 is about 2.9 million RLC, which dwarfs the roughly \$316,000 of firm depth. It should be spread over about fourteen days at roughly 207,000 RLC per day.
- 2 Optimise execution, but expect modest gains.** Use small orders of no more than about 1,000 RLC, paced below 484 RLC per minute and distributed across all six venues by depth, fees, and dealer positioning. Execution friction is only about 230 basis points.
- 3 The principal cost is the holding period.** The all-in cost of 6% to 27% depends on RLC's price path, and the dominant component is the asset's own decline over the unavoidable two weeks.
- 4 Monitor the price trend, not the bid-ask spread.** The decision variables are the number of days held and the price path. The program should be reassessed if the decline re-accelerates.

APPENDIX

How the result was derived

This appendix presents the sequence of models, from M0 to M10, together with the methodology behind the figures.

The model ladder relaxes one assumption at a time and measures the cost that each one adds



How to read it

Phase I (models M0 to M4)

begins from a frictionless benchmark and then introduces finite, firm, and fee-adjusted depth in turn. The amount that can be sold at a single instant falls to about one third of the target.

Phase II (models M5 to M10)

adds time, the behaviour of market makers, the recovery of price impact, the price trend, and cross-venue coupling. Together these produce the final cost band.

The analysis rests on public order-book data and is validated from end to end

The data

The dataset comprises seven venues, about 1.89 million order-book snapshots, 19 million Level-2 updates, and 36,689 trades over roughly 4.7 days, with Binance's trade record backfilled from the public archive. The reconstructed order book matches the independently recorded one to between 98.5% and 100%.

The corrections

The Coinbase USD and USDC feeds were byte-identical and were counted once. Bitget was excluded because its feed re-synchronised after errors, and the Kucoin BTC conversion pair was excluded. Each correction makes the figures more conservative.

The validation

Every estimator verifies itself and recovers known targets on synthetic data; the full analysis passes 28 of 28 end-to-end checks; and an independent square-root impact law cross-checks the cost.